



Mispriced Tail Risk and Optimal Exit Boundaries

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11/23/25

1 Abstract

This paper delves into methodology for assessing mispricings in options contracts through tail risks as well as determining optimal exit strategies based on regime parameters and jump diffusion modeling. We integrate hidden Markov regime detection, jump-diffusion modeling, and optimal stopping to find mispriced contracts and determine when optimal early exercise boundaries to maximize portfolio return rates allowing for efficient capital deployment. A market inefficiency score is used to quantify arbitrage and volatility occurrences across the implied volatility surface. We solve the Hamilton-Jacobi-Bellman partial differential equation to derive optimal exercise boundaries for differing market regimes, moneyness, and times to expiration. Two hypotheses test whether markets misprice tail risk and whether dynamic exit strategies outperform holding to expiration. Validation through Monte Carlo simulation and Fokker-Planck analysis aim to confirm that optimal boundaries significantly improve risk adjusted returns allowing for earlier capital reallocation. This research will provide a foundation for systematic options trading strategies dependent on regime dynamics and jump diffusion along with portfolio efficiency.

2 Introduction

The options market trades more than \$1 trillion dollars each day. Despite this, mispricing exists in the market providing opportunities to informed traders. Traditional pricing models assume constant volatility and drift while regime detection and price jumps are ignored.

Furthermore, American options possess greater potential inefficiencies as traders have the option to exercise contracts early. While theoretically exercising early is suboptimal, portfolios face limited capital constraints which could be allocated to more optimal positions. This poses the question, can tail risk analysis reveal systematic mispricings in options contracts and can mispricings lead to a dynamic exit strategy which optimizes return rates on American options within capital-constrained portfolios?

This research directly benefits AlgoGators by providing a framework for options trading which is a currently underutilized asset class in the fund's portfolio. By deriving optimal exit opportunities, the research provides a foundation for portfolio level strategies applicable to the fund. Additionally, the use of regime detection, jump diffusion, and dynamic exit boundaries offers tools for risk management and capital efficiency. Through finding inefficient contracts, the work done will set a foundation for future strategies utilized by the club.

To address this research question, we look deploy the following hypotheses:

1. Since real portfolios have capital constraints, a dynamic exit strategy from an options contract outperforms the return rate of holding until expiration due to efficient allocation of money.
2. Out of the money, tail options are systematically overpriced, meaning strategies could see profit by selling such contracts.

To test these hypotheses, we employ the following methodology. A Hidden Markov Model (HMM) and Lee Mykland (2008) jump detection testing is deployed on historical training data to identify market regimes and parameters. Option contract data is then interpolated to contract a volatility surface with efficiency scoring. To test hypothesis 1, we find optimal selling boundaries through the Hamilton-Jacobi-Bellman PDE and model option movements with Merton jump diffusion along with Monte Carlo validation to test the return rate of early exercise at the optimal boundaries. To test hypothesis 2, we will use the Breeden Litzenberger methodology to find implied probability distributions and the Fokker-Planck PDE to find realized distributions. By comparing the tails of the distributions, we will determine if implied prices systematically exceed realized.

3 Methodology

(3.1) Data Collection and Filtering

Contract data was sourced from EODHD's option api due to an abundance of historical contract data for an affordable price on both calls and puts with extensive chain coverage including implied volatility. Contract data ranges from November 1st 2023 to December 22, 2024 for tickers SPY, QQQ, IWM, AAPL, MSFT, TSLA, XOM, and JPM. The assets were chosen due to availability of data as well as high liquidity leading to tighter bid-ask spreads. Yahoo Finance historical close prices (non-adjusted) detail underlying asset prices from January 1st 2016 to December 22, 2024. All underlying close prices before the first as of date (November 1st 2023) are used as training data while proceeding prices are used for testing purposes which ensures ample training data.

Data was filtered to ensure quality and relevance of contracts:

- Minimum Bid Price (\$0.05) - eliminate penny options with minimal liquidity.
- Minimum Mid-Price (\$0.10) - ensures economically meaningful option values
- Maximum Bid-Ask Spread (50%) - removes illiquid contracts
- Days to Expiration Range (7-730 days) - focuses on intermediate term options, filters out short term, heavily volatile contracts
- Log Moneyness Bounds (-0.4 to 0.22) - ensures relevant strike near the money
- Implied Volatility Range (5%-200%) - removes extreme volatility contracts
- Price $\geq$ 95% Intrinsic Value – enforces no arbitrage, small tolerance for bid ask bounce
- Time Value Sanity Check ($\leq 500\%$ of intrinsic when intrinsic $> \$1$) - removes contracts with implausible time values

All timestamps are converted to time zone naive dates at midnight for consistency. From contracts, days to expiration, moneyness metrics, and Greeks are calculated. Contracts with missing data are removed for high data quality. Moreover, both risk-free rates and implied dividend yields are obtained from Yahoo Finance using $^{\wedge}\text{IRX}$ and forward dividend yields respectively. For the following sections, comprehensive sensitivity analysis, fallback mechanisms, and validation procedures are detailed in Appendices A-G.

(3.2) Hidden Markov Model Regime Detection:

A Hidden Markov Model is used to identify historical regimes and regime parameters. Four regimes are selected to balance model complexity with interpretability and capture essential market states. Additionally, they separate drift and volatility, which is necessary in options where volatility has a substantial impact on returns. Log returns are used to ensure symmetric, time additive returns. Realized volatility is computed with a window length given by the sensitivity analysis detailed in appendix A.

To initialize the model, k-means is used to create a temporary assignment of training days into one of four clusters. This is done with scikit-learn's default parameters and a random state of 42 for reproducible regime detection. Next, the Baum-Welch algorithm is deployed which iteratively refines regimes, maximizing the likelihood that the observed data could have occurred using the convergence criteria detailed in Appendix A. Diagonal covariance is used to prevent overfitting and ensure stability. From this, we calculate regime parameters, bell curves, and transition probabilities, standardized to zero mean and unit variance.

To run the model on the testing period, we multiply our confidence in the previous day's regime by the probability that today's observations fit into regime bell curves, by the transition probability.

The HMM model outputs:

- Regime specific parameters - Drift (μ), volatility (σ)
- Transition matrix (4x4) - Probability matrix for regime switching
- Historical regime sequence – Daily regime classifications of training period
- Current regime – Identification of the testing day's regime

Regime separation allows future pricing modeling to be based on current market conditions.

(3.3) Jump detection:

The corrected Lee Mykland (2008) jump detection test is used to identify historical jumps crucial for option pricing. Significance level $\alpha = 0.01$ selected through sensitivity analysis in Appendix A. A 22-day (1 month) rolling window bipower variation captures recent volatility to find critical movement values from Gumbel distribution approximation.

The Lee-Mykland test is run through the entire training period divided by asset and regime to identify movements that qualify as jumps. From this, we compute Jump intensity (λ), annual frequency of jumps, and Jump Size Distribution, log normal distribution. A minimum of 2 jumps and 50 observed days within a regime are required to estimate regime specific jump

parameters to ensure reliability on the consistency of jumps. If there is too limited regime specific data for testing, testing is done per without regard for regime. Fallback mechanisms for parameter estimation in Appendix B.

(3.4) Implied Volatility Surface Construction

Volatility surface grid specifications in Appendix A. To account for limited data, we interpolate contracts given the following no arbitrage constraints where we calculate synthetic prices from implied volatility, using Black-Scholes, apply constraints, and convert back to implied volatility. Dividend yield specifics and sources in Appendix C.

No arbitrage constraints:

- Put Call Parity: Maintains option relationships while allowing for bid ask spread:

$$C - P = S \times e^{-qT} - K \times e^{-rT}$$

- Butterfly Spreads: Ensures the volatility curve shape is smooth to avoid static arbitrage:

$$\text{Calls: } C(K-\Delta) - 2C(K) + C(K+\Delta) \geq 0$$

$$\text{Puts: } P(K+\Delta) - 2P(K) + P(K-\Delta) \geq 0$$

- Monotonicity: Total variance $\sigma^2(T)$ doesn't decrease ensuring normal term structure:

$$\sigma^2(T_2) \times T_2 \geq \sigma^2(T_1) \times T_1 \text{ for all } T_2 > T_1$$

A Piecewise Cubic Hermite Interpolation Polynomial (PCHIP) is used to preserve volatility, while linear interpolation of total variance ($\sigma^2 T$) ensures monotonicity. To estimate volatility for synthetic points, linear interpolation is used ($\text{volatility}^2 \times \text{time}$) as it naturally accounts for how variance builds over time. A minimum implied volatility floor of 3% is used as a safeguard against unrealistically low volatility estimates. This full methodology produces a smooth arbitrage free volatility surface. Formula derivations for no-arbitrage constraints in Appendix D.

(3.5) Market Inefficiency Score (MIS)

$$MIS = w_1 z_1 + w_2 z_2 + w_3 z_3 + w_4 z_4$$

z_i = standardized component (butterfly, PCP, skew, term structure)

$$w_i = \text{inverse variance weight } (1/\sigma_i^2) / \sum (1/\sigma_j^2)$$

We compute a Market Inefficiency Score (MIS) to determine inefficiencies within the option volatility plane by combining four arbitrage signals into a single signal. We deploy inverse variance weighting, detailed in Appendix D, to give more influence to constant signals and reduce noise from volatile ones. The signals are as follows:

- Butterfly Violation: Identifies arbitrage opportunities on the volatility smile

$$\text{Magnitude of } C(K-\Delta) - 2C(K) + C(K+\Delta) < 0$$

- Put Call Parity Deviation: Identifies basic arbitrage violations between puts and calls
Absolute deviation from $C - P = S - K \times e^{-rT}$
- Volatility Skew Anomaly: Identifies unusual skew patterns that implies a misprices tail risk. It is calculated by taking the slope of implied volatility versus log moneyness, smoothing it with a Gaussian filter, then measuring the absolute difference between the smoothed slope and its historical median.
- Term Structure Anomaly: Identifies inverted term structures implying market uncertainty. It is calculated by finding where volatility decreases with time to expiration, smoothing these slopes with a Gaussian filter, and measuring the size of these negative slopes.

MIS scores are then standardized by Z-scores from training data means and standard deviations to eliminate discrepancies in score ranges between assets. The 95th percentile of training MIS scores are used to select inefficient contracts to ensure meaningful contracts after bid-ask spreads with a sufficient sample size.

(3.6) Hamilton-Jacobi-Bellman PDE Solution

To address hypothesis 1, we aim to determine the optimal boundary between the continuation and exercise periods of options contracts for American-style option positions. To do so, we solve the Hamilton-Jacobi-Bellman variational inequality on each asset separately for puts and calls. This PDE determines the valuation of contracts for the same underlying assets. While solving, we will keep track of the underlying asset price in which it becomes profitable to exercise at the current time stamp to construct optimal boundaries. A PDE is required as there are two unknowns at each time step: the option value and whether or not it is optimal to exercise at the time stamp. For American options the terminal condition at maturity states that the option's value equals the immediate exercise payoff. Thus, to solve, we start at maturity and work backwards since we can use the boundary condition of the immediate exercise payoff. For each underlying asset, we select one near-the-money put and one call with 30-180 days to expiration ($|\log \text{moneyness}| < 0.05$), as these provide the clearest test cases for early exercise optimality, balancing time value and intrinsic value. Additionally, volatility inputs are interpolated from the arbitrage free implied volatility surfaces constructed in Section 3.4. Under risk neutrality, the asset's drift is adjusted for dividends and jump parameters. To do so, price movement between each time step is defined through the following infinitesimal generator ($\mathcal{L}$):

$$\mathcal{L}V = (r - q - \lambda\kappa)S\partial V/\partial S + \frac{1}{2}\sigma^2 S^2 \partial^2 V/\partial S^2 - rV + \lambda \int [V(S(1+y), t) - V(S, t)] \cdot J(y) dy \text{ where } \sigma, \lambda, \mu_J, \text{ and } \sigma_J \text{ are regime-specific parameters obtained from the HMM model (Section 3.2) and jump detection analysis (Section 3.3). } \kappa = e^{(\mu_J + \sigma_J^2/2)} - 1 \text{ is the jump compensator that}$$

adjusts the drift for expected jump impacts. $J(y) \sim \text{Lognormal}(\mu_J, \sigma_J^2)$ is the jump size density. The jump integral is discretized using 100 points with $\pm 4\sigma$ truncation. For each grid point, jumped prices $S \cdot y$ are interpolated linearly between adjacent grid points.

At each time stamp, an option value is the maximum of exercising at the current time stamp and value for holding until the next node. This ensures that the option value always dominates the immediate exercise value. In the continuation region, the value function $V(S,t)$ satisfies:

$\partial V / \partial t + \mathcal{L}V = 0$ subject to $V(S,t) \geq G(S,t)$, where $G(S,t) = \max(\phi(S - K), 0)$ represents the immediate exercise payoff, with $\phi = +1$ for calls, -1 for puts.

The PDE is solved on a spatial grid of 800 logarithmically spaced points spanning $[0.1S_0, 2S_0]$ and 500 temporal steps from present to maturity. Numerical stability measures for PDE solution in Appendix E. The solution proceeds backward from maturity using an implicit Euler finite difference scheme for the diffusion-drift operator with upwind discretization for the drift term, while the jump term is integrated explicitly at each time step. Regime assignments use exact date matching with 5-day fallback tolerance for holidays and weekends. Fallback mechanisms for regime assignment detailed in Appendix B.

The optimal exercise boundary occurs where the option's value equals its intrinsic value within a relative tolerance of 10^{-7} . Boundary conditions are imposed at the spatial grid extremes: for puts, deep in-the-money ($S \rightarrow 0$) the value approaches $K - S$, while deep out-of-the-money ($S \rightarrow \infty$) the value approaches zero. For calls, the conditions are reversed: deep out-of-the-money ($S \rightarrow 0$) yields zero value, while deep in-the-money ($S \rightarrow \infty$) the value approaches $S - K \times e^{-rT}$. We define the optimal boundaries as the threshold at each time where the option value equals the exercise payoff. The PDE is solved for each asset using the regime prevailing on the contract observation date. This produces regime specific optimal exercise boundaries dependent upon moneyness and time until expiration.

(3.7) Monte Carlo Validation

With the optimal boundary found, we intend to test Hypothesis 1, whether a portfolio with limited capital could utilize early exercise at the optimal boundary to see improved annualized returns, thus allowing for effective capital management. To test this, we use Monte Carlo Simulation with 10,000 simulated paths per each of the contracts analyzed in the HJB solution. Paths are generated through Merton jump diffusion with regime-specific parameters from the regime prevailing on the contract observation date. Contracts for validation are selected from near-the-money options ($|\log \text{moneyness}| < 0.05$) with 30-180 days to expiration that have corresponding HJB optimal boundary solutions. Near-ATM options are selected as they provide the clearest test cases for early exercise decisions,

balancing time value, and intrinsic value. The simulation is discretized into daily time steps. Jump simulation uses lognormal jump sizes and drift compensation for expected jumps ($r - q - \lambda\kappa - \frac{1}{2}\sigma^2$).

The optimal strategy exercises a put when the underlying price falls to or below the boundary, or a call when the price rises to or above the boundary. Returns are calculated as $(\text{Payoff} - \text{Cost})/\text{Cost}$ and converted to annualized log returns: $\ln(1 + \text{ROI})/T_{\text{hold}}$, where T_{hold} is the actual holding period. The optimal strategy's annualized returns are compared with those of always holding until expiration. A minimum hold period of 7 days as well as safety clipping was used to prevent unrealistic annualization when holding periods are too short. Specifics can be found in appendix E. Statistical testing utilizes a paired t-test on annualized log returns ($\alpha = 0.05$) to assess whether the difference in strategies is statistically significant. Additionally, we calculate the improvement rate as the proportion of paths in which the optimal strategy beats the baseline across all the different scenarios or samples tested. This validation framework intends to provide rigorous testing for Hypothesis 1.

3.8) Fokker-Planck Tail Risk Analysis with MIS Integration

Next, we test Hypothesis 2 by comparing realized and implied distributions. We focus on the top 5 dates per symbol with the highest concentration of inefficient contracts (MIS > 95th percentile) as identified in Section 3.5. This approach focuses computational resources on contracts where Step 5 identified potential mispricings, allowing us to test whether these inefficiencies come from systematic tail risk mispricing. Only contracts with expiration dates within 5 days of available IV surface grid points are analyzed to ensure accurate implied distribution extraction. To find the realized distribution for risk neutral constraints, we solve the Fokker-Planck equation using regime-specific volatility σ and jump parameters (λ, μ_J, σ_J), with risk-neutral drift $\mu = r - q - \lambda\kappa$:

$$\partial p / \partial t = -\partial / \partial S [\mu S \cdot p] + \frac{1}{2} \partial^2 / \partial S^2 [\sigma^2 S^2 \cdot p] + \lambda J[p(S/y, t) \cdot (1/y) \cdot J(y) - p(S, t)] dy$$

The spatial grid is constructed from the IV surface moneyness grid ($S = S_0 \cdot e^m$) with the number of points depending on surface density and 500 time steps, to ensure alignment between distributions. The initial condition is a Gaussian distribution centered at S_0 with standard deviation $\sigma S_0 \sqrt{\min(T, 0.04)}$ to ensure proper numerical diffusion for short horizons. Spatial derivatives were discretized using finite differences. The ODE is solved using SciPy's odeint solver (rtol=1e-6, atol=1e-8) with absorbing boundary conditions.

To calculate implied distribution, we deploy the Breeden-Litzenberger methodology which states that the second derivative of call prices (with a small adjustment for interest rates) is the probability density function. We extract the risk-neutral density via:

$$p_{\text{implied}}(K) = e^{rT} \cdot \partial^2 C / \partial K^2$$

Using implied volatilities interpolated from the arbitrage-free surface (Section 3.4), we compute Black-Scholes call prices across strikes, then apply spline differentiation to get the smooth probability densities ($\partial^2 C / \partial K^2$).

Importantly, both distributions are risk neutral as Fokker-Planck uses risk neutral drift, and the Breeden-Litzenberger methodology directly extracts risk neutral density implied by option prices. For comparison, the tail regions are as follows: Left ($S < 0.9S_0$ for OTM puts), Right ($S > 1.1S_0$ for OTM calls) to focus on economically relevant extreme moves. To assess overpricing, we compute tail mass ratios, KS statistics, and volatility premiums to quantify the statistical and economic significance of mispricing. Validation for this testing can be found in appendix F.

4 Results

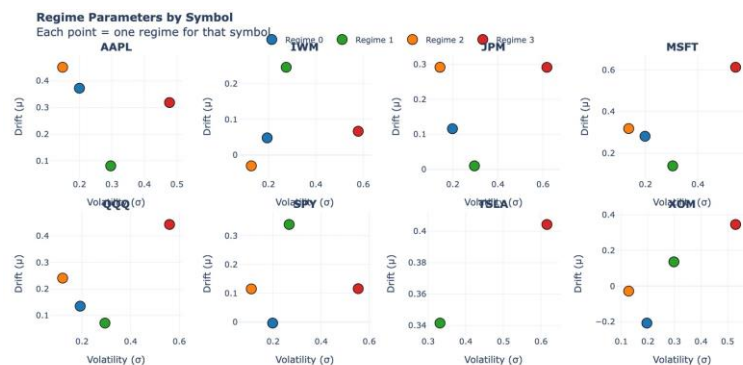
Our results gave strong support for both of our research hypotheses with significant findings.

4.1 Data extraction and quality filtering

From an initial extraction of 9,336,744 option contracts, the quality filtering kept 3,712,738 high quality contracts (39.8%) across eight highly liquid underlying assets (SPY, QQQ, IWM, AAPL, MSFT, TSLA, XOM, JPM) from November 1, 2023 to December 20, 2024. Of the retained data, there are 1,868,325 calls and 1,844,413 puts with an average implied volatility of 26.1% and average bid-ask spread of 4.19%. Filtering ensured the final dataset focused on intermediate term contracts (7–730 DTE), near-the-money strikes (log moneyness -0.40 to $+0.22$), and satisfied no arbitrage and liquidity criteria including minimum pricing, maximum spread $\leq 50\%$, price $\geq 95\%$ of intrinsic value, and extreme time-value removal. Moneyness was tightly centered around ATM levels (call mean $\log(K/S) = -0.067$, put mean $= -0.049$). This provided clean and relevant contract data ensuring the significance of later results.

4.2 Market Regime Detection and Jump Parameters

The Hidden Markov Model was calibrated using sensitivity analysis which found the 60 day window to be most optimal. The HMM identified four distinct market regimes across the assets with meaningful parameter differentiation. This parameter separation, which is shown per asset by the above graph, confirms the model's ability to identify market states. The regime structure



demonstrated strong persistence with diagonal transition probabilities exceeding 0.98 for all 8 assets. This supports the stability of identified regimes as shown in Appendix E.

Similarly, jump detection analysis found 96 significant price jumps across all assets during the training period. The regime specific jump intensities ranged from 0.56 to 1.61 jumps per year.

4.3 Dynamic Exit Strategy Performance (Hypothesis 1)

Monte Carlo validation was run 16 times to account for assets and contract type and found significantly improved return rates for the optimal exercise strategy. The analysis revealed that the optimal strategy produced annualized returns that were statistically significant ($p < 0.05$) for all tests. The optimal strategy saw an average annualized return improvement of 133.55% and exercised early 55.5% of the time. The magnitude of these improvements, though misleading as discussed in section 5.3, demonstrates greater capital efficiency from earlier exercise compensates for the sacrificed time value.

As expected, the hold-to-expiration strategy saw higher total returns per trade. The optimal strategy had lower total returns in 15 of 16 cases (optimal outperformed only for QQQ puts). The optimal strategy saw +62.83% annualized returns on average. This outcome confirms the theoretical value of time premium which is fully captured only at expiration. This further validates the findings of the paper as modeling demonstrates known market behavior.

4.4 Tail Risk Mispricing Analysis (Hypothesis 2)

The Fokker-Planck and Breeden-Litzenberger methodology was used to analyze 40 high-MIS contracts across the eight symbols in order to test systematic tail risk mispricing. The analysis passed all six validation checks as detailed in Appendix F, confirming the integrity of both the realized and implied probability distributions.

The results showed a consistent tail risk overpricing. For out-of-the-money (OTM) puts, 35 of the 40 contracts analyzed (87.5%) were overpriced. For OTM calls, 29 of the 40 contracts (72.5%) were overpriced. The Kolmogorov-Smirnov statistics quantifies the difference between the implied and realized distributions. In analysis, these values ranged from 0.1976 to 0.3537, which confirms the significance of the distribution of misalignments.

5 Discussion

The results provide strong support for both research hypotheses, concluding that regime dependent, jump-diffusion models can be used to identify systematic mispricings and that dynamic exit strategies significantly enhance capital efficiency.

5.1 Interpretation of Regime and Jump Dynamics

The regime classification was powerful in capturing distinct market states, with parameters being backed by economic intuition and accurate condition labeling. The positive drift parameters seen in Bear regimes can likely be attributed to the model's 60-day lookback window. Given a significant look back, prolonged bear markets may capture rebound rallies which could be reflected in regime parameters. Assuming this statement to be true, positive drift could be interpreted as a reflection of the chosen calibration period, effectively identifying recovering states. Similarly, the jump parameters further validate the model's design as the highest volatility regimes corresponded with the highest jump intensity in the testing period. This confirms that the model is correctly associating high-volatility environments with a greater frequency of jumps. This strength of the model is crucial for accurately pricing tail risks and stock movements.

5.2 Hypothesis 2: Systematic Tail Risk Mispricing

The analysis of 40 high-MIS contracts confirms that markets systematically overprice tail risk. The finding that 87.5% of OTM puts and 72.5% of OTM calls were overpriced points to a significant overestimation of future volatility and extreme price movements. This systematic mispricing identifies a consistent opportunity for future strategies that can more accurately model realized price distributions.

5.3 Hypothesis 1: Capital Efficiency of Dynamic Exit

The Monte Carlo validation found definitive support for the optimal exit strategy. Specifically, the contrast between total return per trade and annualized return rate gives context to the behavior. As expected, the hold to expiration strategy generated higher total returns in most cases (15 of 16 contracts) as it captures the option's full time value which decays to zero at expiration. However, this metric is less relevant for a capital constrained portfolio because it does not take the opportunity cost of deployed capital into account. Thus, by assessing the annualized return, which accounts for the holding period, we get a stronger assessment of a constrained portfolio's potential returns. The optimal strategy's substantially better return rate (averaging +75.24% for puts and +191.86% for calls) demonstrates early exercising at optimal boundaries to allow for earlier capital recycling despite diminished raw returns.

While the significantly higher annualized gains do provide ample support for our hypothesis, it is important to address the context for the magnitude of these numbers. The reported $\approx 130\%$ annualized improvements represent a theoretical maximum for ideal market conditions where there is frictionless trading and a continuous supply of identical mispriced. Real world transaction costs, bid-ask spreads, and commissions could further lessen profits from trading more frequently. Additionally, in practice, there are cash drags between identified opportunities. Therefore, the correct interpretation of the magnitude is that the optimal exercise strategy has the potential for $\approx 130\%$ annualized improvements over the holding strategy.

5.4 Limitations and Future Research

This study has several limitations. Even though the 60-day window is supported by sensitivity analysis, it may not capture longer term economic cycles. While the idea of regime persistence is supported by the high transition probabilities, it may not hold during periods of sudden regime shift. Furthermore, the HJB solutions assume perfect liquidity with the ability to trade at the continuous boundary. While highly liquid assets were chosen to address this, it still may not be completely possible in practice. Future research could delve into implementing live trading for optimal exit strategy. This could be done by analyzing factor regression on optimal boundaries using components such as moneyness, time to expiration, and the volatility regime to bypass complex PDE calculations.

6 Conclusion

In total, this research supports the claim that the options markets systematically misprice tail risk and that dynamic exit strategies can be used to improve capital efficiency. The regime specific approach was successful in finding mispriced contracts. Similarly, the optimal exercise boundaries found 130%+ annualized return improvements through faster capital reallocation.

The findings in this paper provide AlgoGators with the foundation to develop options trading strategies. The methodology for detecting volatility surface inefficiencies and determining optimal exit points can be used across market regimes and other asset classes. This could give the fund a researched approach to options trading.

7 References

- Alali, G. L. O. M. (n.d.). Optimal stopping, free boundary, and American option in a jump-diffusion model. ResearchGate. Retrieved from https://www.researchgate.net/publication/227230520_Optimal_Stopping_Free_Boundary_and_American_Option_in_a_Jump-Diffusion_Model
- Bolia, N., & Juneja, S. (2005). *Monte Carlo methods for pricing financial options*. Sādhana, 30(2–3), 347–385. https://web.iitd.ac.in/~nomesh/pubs/survey_MC_methods_options.pdf
- Davis, M. H. A. (n.d.). *Local volatility, stochastic volatility and jump-diffusion models*. Columbia University. Retrieved from <http://www.columbia.edu/~mh2078/ContinuousFE/LocalStochasticJumps.pdf>
- Hanson, F. B. (n.d.). *Stochastic analysis of jump-diffusions for financial log-return processes* (corrected). University of Illinois Chicago. Retrieved from <https://www.math.uic.edu/~hanson/pub/KU02/ku02hwfmcor.pdf>
- Jia, Q. (2009). Pricing American options using Monte Carlo methods (Master's thesis, Uppsala University). U.U.D.M. Project Report 2009:8. <https://www.diva-portal.org/smash/get/diva2:301061/FULLTEXT01.pdf>

- Wang, F. Y. (2010). *Fokker-Planck equations for a free energy functional or Markov process on a graph* (CAM Report 10-45). University of California, Los Angeles, Department of Mathematics. Retrieved from <https://www.math.ucla.edu/camreport/cam10-45.pdf>
- Yang, P., & Zhang, Y. (2014). *Jump detection with wavelets for high-frequency financial time series*. ResearchGate. Retrieved from https://www.researchgate.net/publication/268152725_Jump_detection_with_wavelets_for_high-frequency_financial_time_series
- Zhang, D., & Melnik, R. V. N. (2007). *Monte-Carlo simulations of the first passage time for multivariate jump-diffusion processes in financial applications* (arXiv:cs/0702164). arXiv. Retrieved from <https://arxiv.org/abs/cs/0702164>

8 Appendices

APPENDIX A: SENSITIVITY ANALYSIS

HMM Convergence Criteria:

- Maximum iterations: 1000
- Convergence tolerance: 1e-6

HMM Window Length Sensitivity:

- Tested windows: [20, 60, 120, 252] days
- The selection criterion was to choose the longest window duration under 100 days to prioritize model stability while maintaining relevance to current market conditions.
- Optimal typically: 60-120 days

Jump Detection Alpha Sensitivity:

- Tested: $\alpha = [0.001, 0.005, 0.01]$
- Selected: $\alpha = 0.01$ (detects ~1-2% of returns as jumps)
- Balance between sensitivity and false positives

Jump integral discretization

- Jump integral: Discretize log-jumps ($\mu_J \pm 4 \sigma_J$, 50 points)
- ($\mu_J \pm 4 \sigma_J$, 50 points)

Monte Carlo Safety Clipping:

- Raw returns clipped: [-95%, +300%]
- Annualized log returns clipped: [-5.0, +5.3]

- Minimum holding period: 7 days
- Prevents extreme annualization artifacts

Monte Carlo Discretized to Daily Time Steps:

- Discretize to $\max(252T, 50)$ steps
- Provides daily resolution with minimum stability
- Balance between accuracy and computation.

Grid Specifications:

- IV Surface (3.4): 101-point moneyness grid (-0.5 to 0.5), 14 DTE expiries
- HJB PDE (3.6): 800 spatial points ($0.1S_0$ to $2.0S_0$), 500 temporal steps
- Fokker-Planck (3.8): Spatial grid derived from IV surface moneyness (101 points from MONEYNESS_GRID), extended to 200 points if needed for stability; 500 time steps

Parameter Sensitivities:

- HMM Window Length (3.2): Tested [20, 60, 120, 252] days, optimal 60-120 days
- Jump Detection Alpha (3.3): Tested $\alpha = [0.001, 0.005, 0.01]$, selected $\alpha = 0.01$
- MIS Threshold (3.5): Tested [90th, 95th, 99th] percentiles, selected 95th percentile

IV Slice at Closest DTE:

- IV slice at closest DTE (± 5 days tolerance)

Spline smooth:

- Spline smooth ($s=0.01$, $k=3$)

APPENDIX B: FALLBACK MECHANISMS

Hierarchical Parameter Fallbacks:

1. Regime Detection: 5-day date matching tolerance
2. Jump Parameters:
 - Level 1: Symbol-regime specific ($n_{\text{jumps}} \geq 2$, $n_{\text{days}} \geq 50$)
 - Level 2: Regime-pooled parameters
 - Level 3: Default values ($\lambda=1.0$, $\mu_J=0.0$, $\sigma_J=0.05$)
3. Spot Prices: 5-day nearest neighbor
4. IV Surface: 5-day DTE matching tolerance

5. HJB Boundary: Conservative fallbacks (80%K puts, 120%K calls)

APPENDIX C: DIVIDEND YIELDS

SPY	QQQ	IWM	AAPL	MSFT	TSLA	XOM	JPM
1.09%	0.48%	1.06%	0.38%	0.77%	0.00%	3.52%	2.01%

Source: Yahoo Finance forward dividend yields; Yields were static

APPENDIX D: FORMULA DERIVATIONS

Breeden-Litzenberger Derivation:

From fundamental theorem: $C = e^{-rT} \int (S-K)^+ f(S) dS$

Differentiate twice: $\partial^2 C / \partial K^2 = e^{-rT} f(K)$

Fokker-Planck for Jump-Diffusion:

From Kolmogorov forward equation with jump term:

$$\partial p / \partial t = -\partial / \partial S [\mu S p] + \frac{1}{2} \partial^2 / \partial S^2 [\sigma^2 S^2 p] + \lambda \int [p(S/y) J(y)/y - p(S)] dy$$

Lee-Mykland Jump Test: Test statistic: $L(i) = |r_i| / \sigma_i$ Critical value: $\beta = (1/\sqrt{2\pi}) * \exp(-0.5(c_n - \beta/c_n)^2)$

APPENDIX E: NUMERICAL STABILITY

IV Surface Construction Stability (3.4):

- Put-call parity enforced with 1% tolerance to accommodate bid-ask spreads and discrete pricing
- Minimum implied volatility floors of 3% prevent unrealistically low volatility estimates
- 5-day DTE matching tolerance for IV surface interpolation
- Normalization after each interpolation step

HJB PDE Stability Measures (3.6):

- Implicit Euler scheme (unconditionally stable)
- Upwind differencing for drift term
- Minimum covariance: 1e-3 in HMM
- Convergence tolerance: 1e-4

Monte Carlo Safety Layers (3.8):

- Return capping at $\pm 300\%$
- Minimum 7-day holding period
- Log-return annualization with caps
- NaN/inf filtering

Fokker-Planck Stability (3.8):

- Absorbing boundary conditions
- Normalization after each step
- Non-negativity enforcement: $p = \max(p, 0)$

APPENDIX F: VALIDATION FRAMEWORK

Six Validation Checks (3.8):

1. PDF Normalization: Integrals = $1.0 \pm 5\%$
2. Mean Consistency: Realized vs theoretical mean < 10% error
3. Variance Comparison: Implied vs realized volatility
4. BS Baseline: Compare with lognormal distribution
5. Vol Premium: Calculate implied - realized vol
6. Skewness: Compare distribution asymmetries

Statistical Tests:

1. KS test for distribution differences
2. Paired t-test for Monte Carlo returns
3. Cohen's d for effect sizes
4. Bootstrap confidence intervals

APPENDIX G: GRAPHS AND VISUALIZATIONS

1. HMM regime switching probability matrix

From / To	Bear/High-Vol	Bull/Low-Vol	Bear/Low-Vol	Bull/High-Vol
Bear/High-Vol	0.9846	0.0078	0.0071	0.0004
Bull/Low-Vol	0.0092	0.9864	0.0000	0.0044
Bear/Low-Vol	0.0092	0.0000	0.9903	0.0005
Bull/High-Vol	0.0000	0.0091	0.0000	0.9909

2. PUTS - Total Return Comparison

Contract	Optimal Return	Hold Return	Difference
SPY PUT	+41.55%	+41.72%	-0.17%
QQQ PUT	+19.11%	+17.63%	+1.48%
IWM PUT	+19.72%	+22.60%	-2.88%
AAPL PUT	+32.33%	+38.82%	-6.50%
MSFT PUT	+36.61%	+39.04%	-2.43%
TSLA PUT	+53.95%	+65.03%	-11.07%
XOM PUT	+22.60%	+24.14%	-1.55%
JPM PUT	+25.06%	+28.74%	-3.68%

Summary: Optimal Outperforms: 1/8 (12.5%) • Avg Difference: -3.35%

3. CALLS - Total Return Comparison

Contract	Optimal Return	Hold Return	Difference
SPY CALL	+87.76%	+127.80%	-40.04%
QQQ CALL	+73.70%	+121.66%	-47.96%
IWM CALL	+73.85%	+119.82%	-45.97%
AAPL CALL	+34.86%	+42.26%	-7.40%
MSFT CALL	+90.20%	+153.21%	-63.01%
TSLA CALL	+96.72%	+160.96%	-64.24%
XOM CALL	+28.18%	+33.37%	-5.19%
JPM CALL	+44.93%	+53.49%	-8.55%

Summary: Optimal Outperforms: 0/8 (0.0%) • Avg Difference: -35.30%

4. PUTS - Annualized Return Comparison

Contract	Optimal Annualized	Hold Annualized	Difference
SPY PUT	-0.84%	-82.77%	+81.93%
QQQ PUT	-51.08%	-88.25%	+37.17%
IWM PUT	-46.92%	-83.23%	+36.31%
AAPL PUT	+4.47%	-80.13%	+84.60%

MSFT PUT	+5.87%	-80.97%	+86.84%
TSLA PUT	+98.08%	-71.20%	+169.29%
XOM PUT	-28.53%	-82.05%	+53.52%
JPM PUT	-30.33%	-82.63%	+52.30%

Summary: Avg Annualized Improvement: +75.24%/year

5. CALLS - Annualized Return Comparison

Contract	Optimal Annualized	Hold Annualized	Difference
SPY CALL	+257.33%	-68.69%	+326.01%
QQQ CALL	+149.35%	-68.14%	+217.49%
IWM CALL	+118.60%	-77.92%	+196.52%
AAPL CALL	-25.97%	-82.50%	+56.53%
MSFT CALL	+269.39%	-74.49%	+343.88%
TSLA CALL	+168.03%	-86.03%	+254.07%
XOM CALL	-31.68%	-83.31%	+51.63%
JPM CALL	+8.85%	-79.89%	+88.74%

Summary: Avg Annualized Improvement: +191.86%/year

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